Temporal bone & skull base

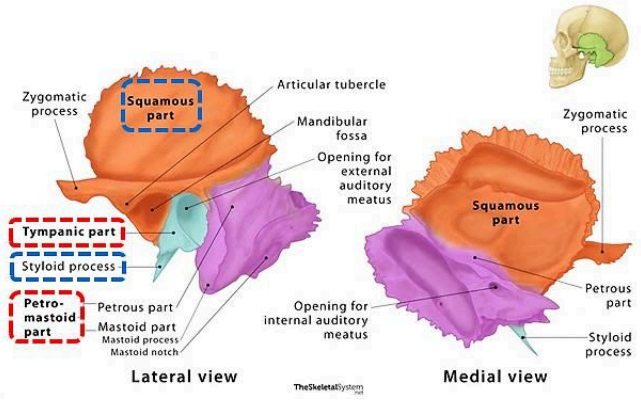
The bony house of the ear: five parts, three petrous surfaces, the foramina of the skull base and the landmarks you drill by.

1. Overview and contents
2. Petrous part: three surfaces
3. Tympanic and mastoid parts, styloid process
4. Foramina of the skull base
5. Pterygopalatine fossa
6. On imaging
7. In theatre

01 Overview and contents

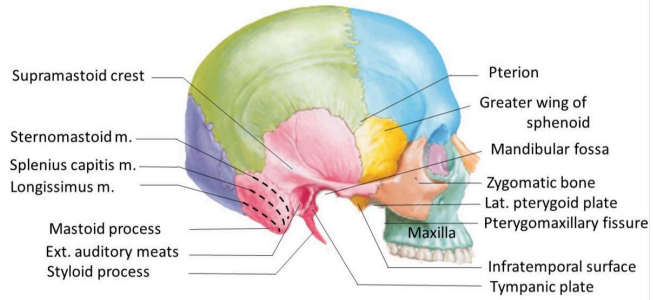
- Forms part of the **base and lateral wall** of the skull; its petrous part is pyramid-shaped. Articulates with 5 bones: occipital, parietal, sphenoid, zygomatic and the mandible (TMJ).
- The **petrous part separates the middle from the posterior cranial fossa**.
- **Houses:** bony EAC, middle ear with the ossicles, inner ear, the fallopian canal (facial nerve), the carotid canal (ICA), and bony cover of the sigmoid sinus and jugular bulb.
- **Five parts:** squamous, petrous, mastoid, tympanic and the styloid process.
- **Squamous part:** lateral wall of the middle cranial fossa and the **roof of the bony EAC**; its zygomatic process roofs the mandibular fossa.

CLINICAL PEARL The ear sits next to the middle and posterior fossae, the jugular bulb, sigmoid sinus, ICA and cranial nerves V–XII. This is why otitis media can produce intracranial complications.



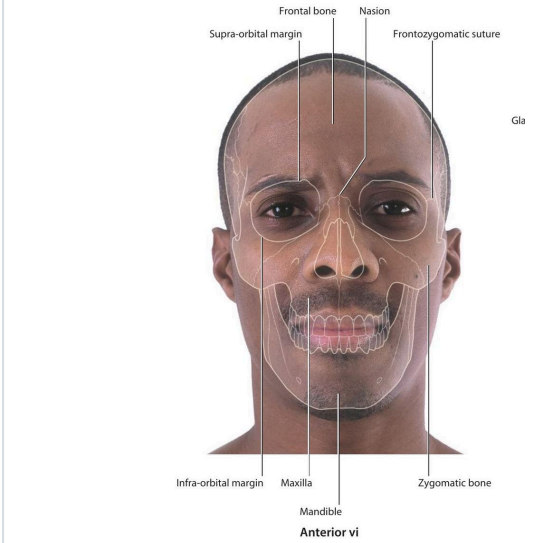
Parts of the temporal bone: lateral and medial views

ENT Steps Anatomy, p. 18



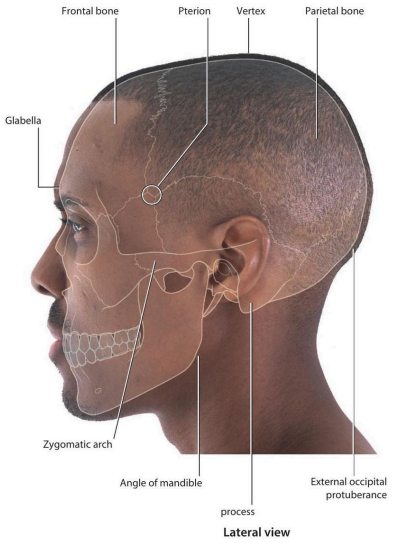
Lateral view of the skull: temporal and infratemporal regions

Dr. Sameh Doss handout, p. 10



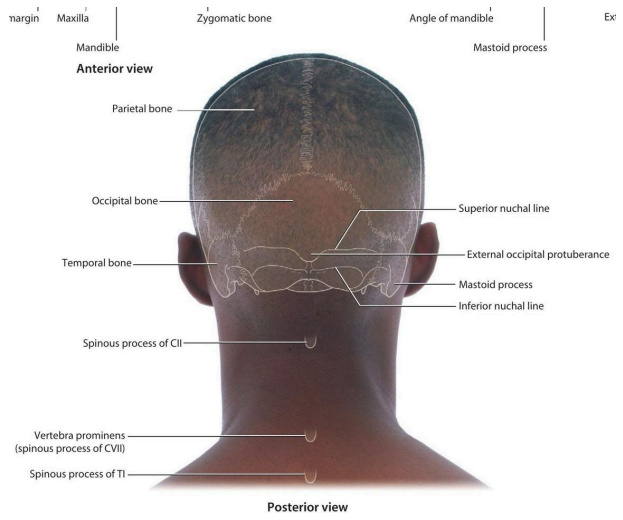
Surface anatomy with bones: anterior view

Gray's Atlas of Anatomy, 2nd ed., p. 476



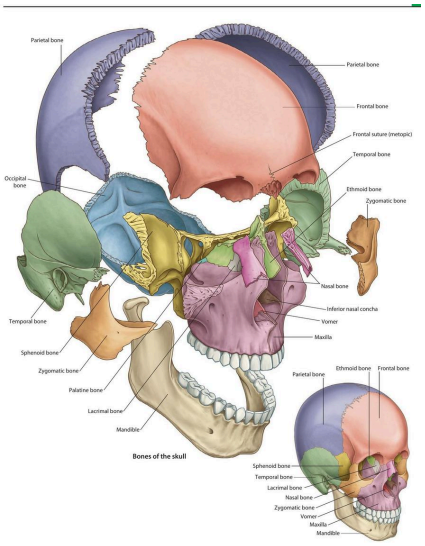
Surface anatomy with bones: lateral view (pterion, mastoid, angle)

Gray's Atlas of Anatomy, 2nd ed., p. 476



Surface anatomy with bones: posterior view

Gray's Atlas of Anatomy, 2nd ed., p. 476



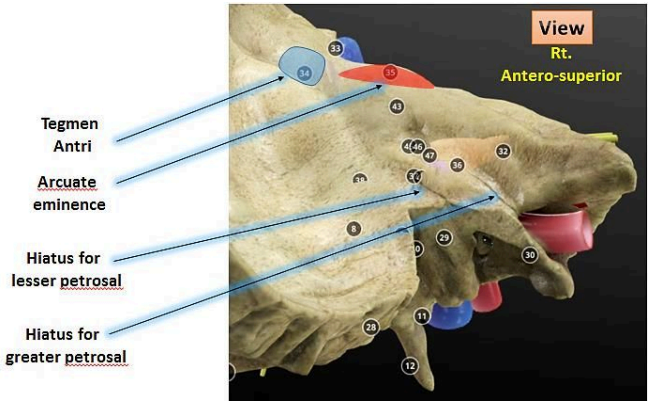
Bones of the skull (exploded view)

Gray's Atlas of Anatomy, 2nd ed., p. 477

02 Petrous part: three surfaces

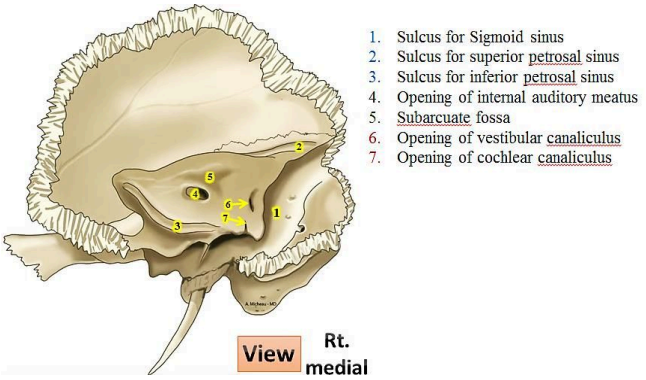
- **Anterior surface** (floor of the middle cranial fossa): **arcuate eminence** of the superior canal medially, **tegmen tympani and tegmen antri** laterally, canal for tensor tympani and the ET in front, **trigeminal impression** at the apex, and hiatuses for the greater and lesser petrosal nerves.
- **Posterior surface** (front of the posterior fossa): **internal acoustic meatus** (VII, VIII, labyrinthine vessels), **subarcuate fossa** superolateral to the IAM, the **operculum** for the vestibular aqueduct inferolateral to it, and grooves for the sigmoid, superior and inferior petrosal sinuses.
- **Inferior surface:** jugular fossa (jugular bulb), lower opening of the **carotid canal** in front of it, separated by a wedge of bone called the **keel (caroticojugular crest)**.
- The **tympanic canaliculus** in the keel transmits **Jacobson's nerve** (tympanic branch of IX). The **cochlear aqueduct** opens at the top of the jugular foramen.
- The **jugular foramen** forms where the petrous and occipital bones meet.

CLINICAL PEARL Petrous apicitis (Gradenigo's syndrome): ear discharge, retro-orbital pain (V at the apex) and lateral rectus palsy (VI in Dorello's canal).



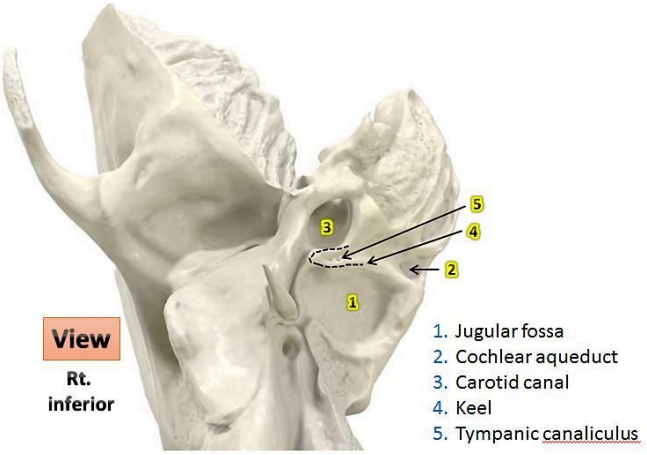
Anterior surface of the petrous bone: tegmen, arcuate eminence, petrosal hiatuses

ENT Steps Anatomy, p. 21



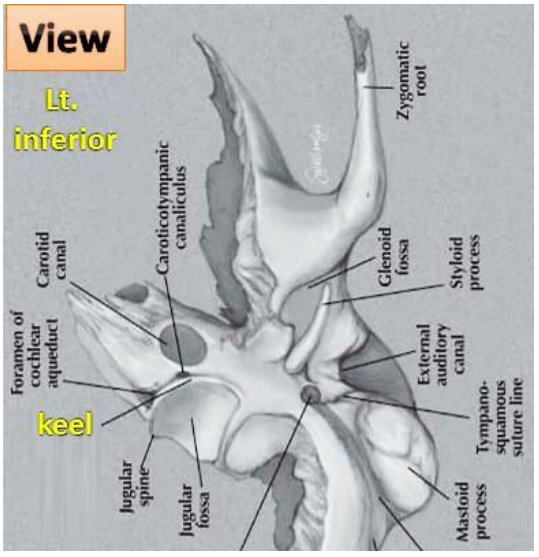
Posterior surface of the petrous bone: IAM, subarcuate fossa, sinus sulci

ENT Steps Anatomy, p. 23



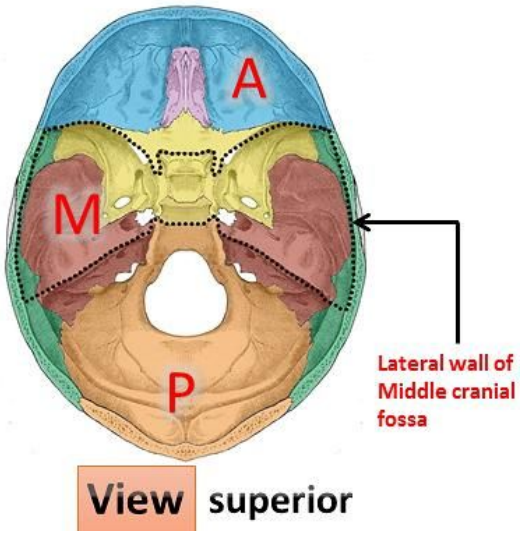
Inferior surface: jugular fossa, carotid canal, cochlear aqueduct

ENT Steps Anatomy, p. 24



Inferior surface in detail with the keel between carotid and jugular

ENT Steps Anatomy, p. 24



Cranial fossae from above: the petrous ridge divides middle from posterior fossa

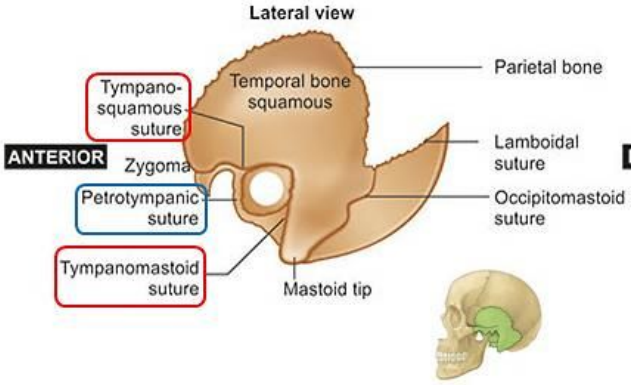
ENT Steps Anatomy, p. 19

03 Tympanic and mastoid parts, styloid process

- **Tympanic part:** forms the anterior wall, floor and posterior wall of the bony EAC, the anterior wall and floor of the middle ear, and the back of the mandibular fossa.
- **Sutures:** in the bony EAC, **tympanosquamous** in front and **tympanomastoid** behind; in the middle ear, the **petrotympanic (Glaserian) fissure**, through which the chorda tympani leaves.
- **Mastoid part:** the tip receives **SCM**; the **digastric groove** gives origin to the posterior belly of digastric and ends in front at the **stylomastoid foramen**; an emissary vein to the sigmoid sinus passes through the mastoid foramen.
- The **mastoid antrum lies deep to Macewen's (suprameatal) triangle**.
- **Styloid process:** in front of the stylomastoid foramen. **Three muscles** (stylohyoid, styloglossus, stylopharyngeus) and **two ligaments** (stylohyoid, stylomandibular).

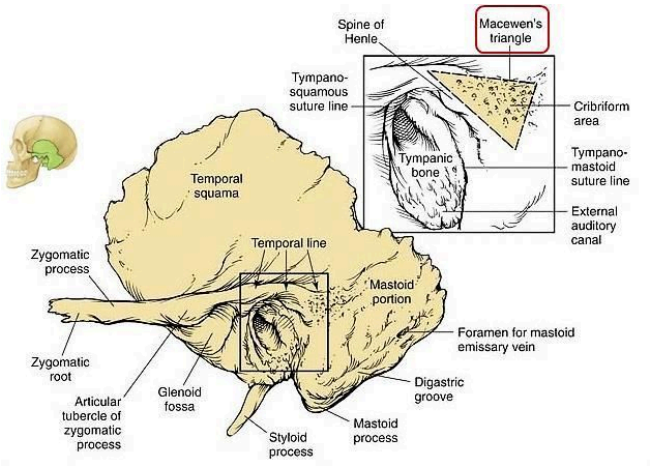
CLINICAL PEARL Stylomandibular ligament separates the parotid from the submandibular gland; an elongated styloid (over 3 cm) causes Eagle's syndrome.

SOURCES DIFFER CHORDA TYMPANI EXIT: ENT Steps says the petrotympanic fissure; Doss says the squamotympanic fissure. Gray's: the anterior canaliculus (canal of Huguier) at the medial end of the petrotympanic fissure.



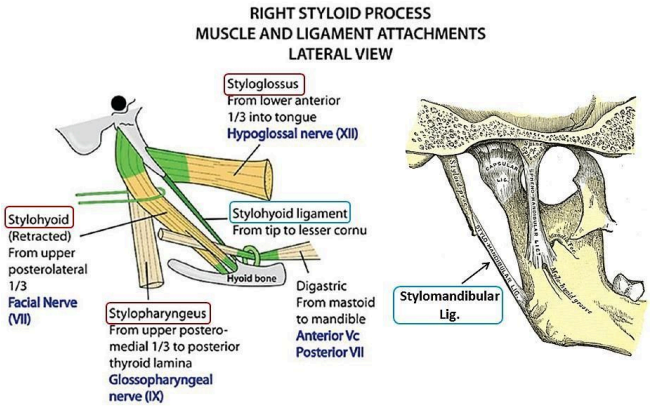
Sutures of the temporal bone, lateral view

ENT Steps Anatomy, p. 25



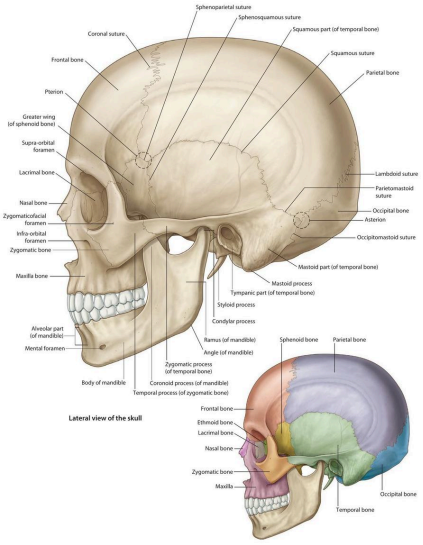
Lateral surface: spine of Henle, Macewen's triangle, mastoid

ENT Steps Anatomy, p. 26



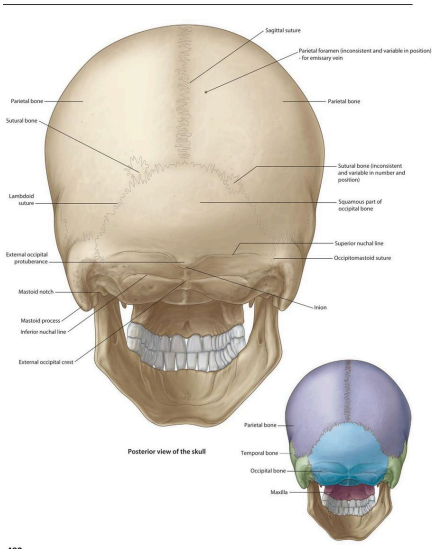
Styloid apparatus: 3 muscles and 2 ligaments

ENT Steps Anatomy, p. 27



Lateral view of the skull: squamous, tympanic and mastoid parts

Gray's Atlas of Anatomy, 2nd ed., p. 480



Posterior view of the skull: mastoid process and notch, occipital bone

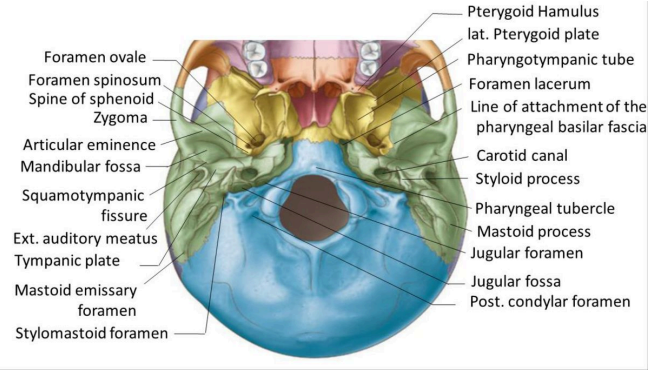
Gray's Atlas of Anatomy, 2nd ed., p. 482

04 Foramina of the skull base

- **Cribriform plate:** olfactory nerves (anterior fossa).
- **Optic canal:** optic nerve and ophthalmic artery.
- **Superior orbital fissure:** III, IV, VI, the three branches of V1 (lacrimal, frontal, nasociliary) and the superior ophthalmic vein.
- **Foramen rotundum:** V2. **Foramen ovale (MALE):** Mandibular nerve, Accessory meningeal artery, Lesser petrosal nerve, Emissary vein. **Foramen spinosum:** middle meningeal artery and the meningeal branch of V3.
- **Foramen lacerum:** filled with cartilage; the ICA crosses its upper part, and the greater petrosal joins the deep petrosal to form the vidian nerve.
- **Internal acoustic meatus:** VII, nervus intermedius, VIII and labyrinthine vessels.
- **Jugular foramen:** IX, X, XI, the IJV (from the sigmoid sinus) and the inferior petrosal sinus. **Hypoglossal canal:** XII. **Stylomastoid foramen:** VII and the stylomastoid artery.

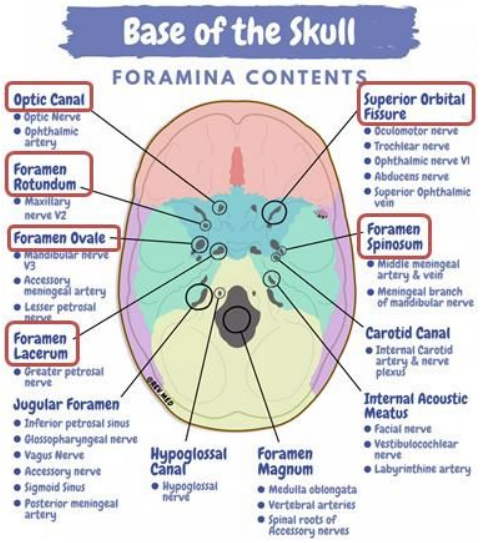
CLINICAL PEARL Jugular foramen syndromes: Vernet = IX, X, XI. Collet–Sicard = IX–XII. Villaret = IX–XII plus the cervical sympathetic chain (Horner's).

SOURCES DIFFER FORAMEN OVALE: Doss adds the motor root of V and puts the middle meningeal vein here ("3 M + ALE"); Gray's sends the middle meningeal vein through the foramen spinosum.



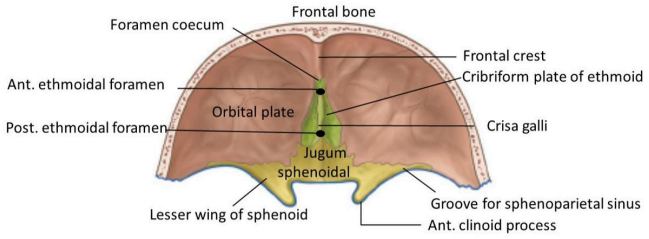
Norma basalis: foramina of the skull base

Dr. Sameh Doss handout, p. 16



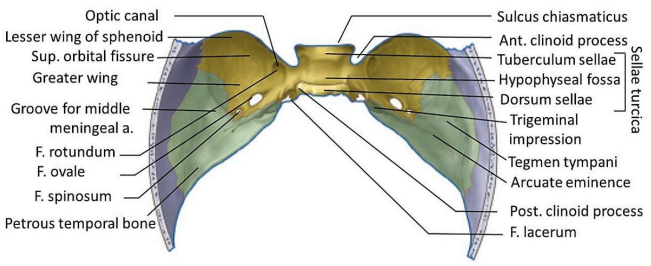
Foramina of the skull base and their contents

ENT Steps Anatomy, p. 152



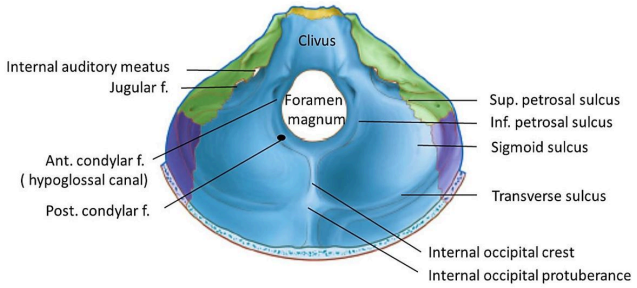
Anterior cranial fossa

Dr. Sameh Doss handout, p. 24



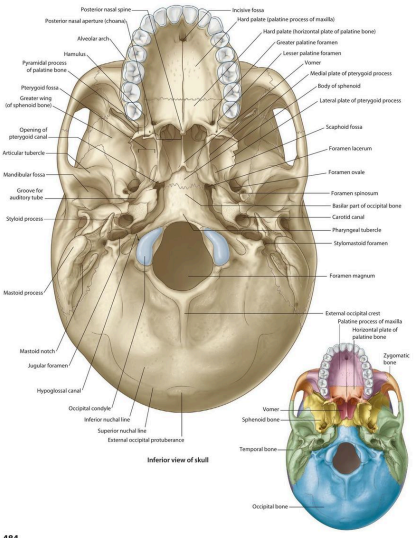
Middle cranial fossa and its foramina

Dr. Sameh Doss handout, p. 25



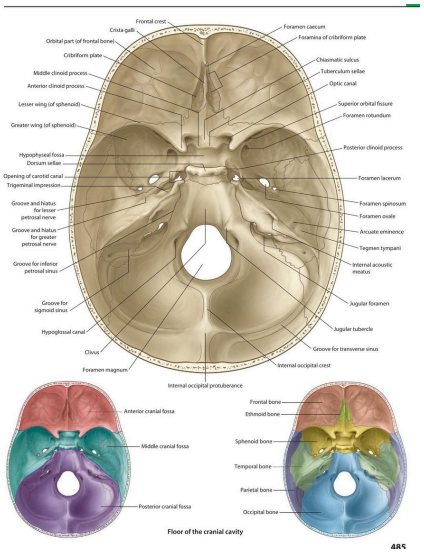
Posterior cranial fossa

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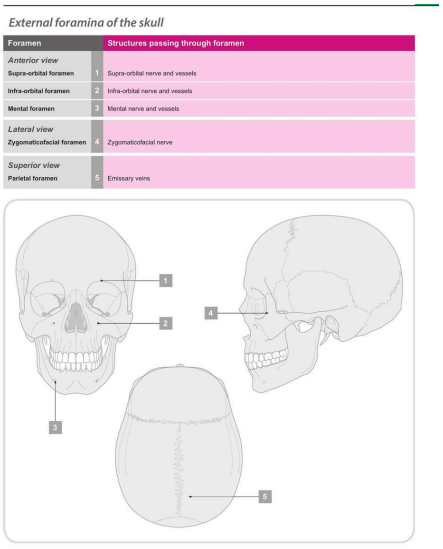
Inferior view of the skull: foramina of the skull base

Gray's Atlas of Anatomy, 2nd ed., p. 484



Floor of the cranial cavity: cranial fossae and foramina

Gray's Atlas of Anatomy, 2nd ed., p. 485



External foramina of the skull

Foramen		Structures passing through foramen
Anterior view		
Supra-orbital foramen	1	Supra-orbital nerve and vessels
Infra-orbital foramen	2	Infra-orbital nerve and vessels
Mental foramen	3	Mental nerve and vessels
Lateral view		
Zygomaticofacial foramen	4	Zygomaticofacial nerve
Superior view		
Parietal foramen	5	Emissary veins

External foramina of the skull and their contents

Gray's Atlas of Anatomy, 2nd ed., p. 587

External foramina of the skull

Foramen		Structures passing through foramen
Inferior view		
Incisive foramina	1	Nasopalatine nerve; sphenopalatine vessels
Greater palatine foramen	2	Greater palatine nerve and vessels
Lesser palatine foramina	3	Lesser palatine nerves and vessels
Pterygoid canal	4	Pterygoid nerve and vessels
Foramen ovale	5	Mandibular nerve [V_3]; lesser petrosal nerve
Foramen spinosum	6	Middle meningeal artery
Foramen lacerum	7	Filled with cartilage
Carotid canal	8	Internal carotid artery and nerve plexus
Foramen magnum	9	Continuation of brain and spinal cord; vertebral arteries and nerve plexuses; anterior spinal artery; posterior spinal arteries; roots of accessory nerve [XI]; meninges
Condylar canal	10	Emissary veins
Hypoglossal canal	11	Hypoglossal nerve [XII] and vessels
Jugular foramen	12	Internal jugular vein; inferior petrosal sinus; glossopharyngeal nerve [IX]; vagus nerve [X]; accessory nerve [XI]
Stylomastoid foramen	13	Facial nerve [VII]

Internal foramina of the skull

Anterior cranial fossa		
Foramen caecum	1	Emissary veins to nasal cavity
Olfactory foramina in cribriform plate	2	Olfactory nerves [I]
Middle cranial fossa		
Optic canal	3	Optic nerve [II]; ophthalmic artery
Superior orbital fissure	4	Oculomotor nerve [III]; trochlear nerve [IV]; ophthalmic division of the trigeminal nerve [V_1]; abducent nerve [VI]; ophthalmic veins
Foramen rotundum	5	Maxillary division of the trigeminal nerve [V_2]
Foramen ovale	6	Mandibular division of the trigeminal nerve [V_3]; lesser petrosal nerve
Foramen spinosum	7	Middle meningeal artery
Hiatus for the greater petrosal nerve	8	Greater petrosal nerve
Hiatus for the lesser petrosal nerve	9	Lesser petrosal nerve
Posterior cranial fossa		
Foramen magnum	10	End of brainstem/beginning of spinal cord; vertebral arteries; spinal roots of the accessory nerve; meninges
Internal acoustic meatus	11	Facial nerve [VII]; vestibulocochlear nerve [VIII]; labyrinthine artery
Jugular foramen	12	Glossopharyngeal nerve [IX]; vagus nerve [X]; accessory nerve [XI]; inferior petrosal sinus, sigmoid sinus (forming internal jugular vein)
Hypoglossal canal	13	Hypoglossal nerve [XII]; meningeal branch of the ascending pharyngeal artery
Condylar canal	14	Emissary vein

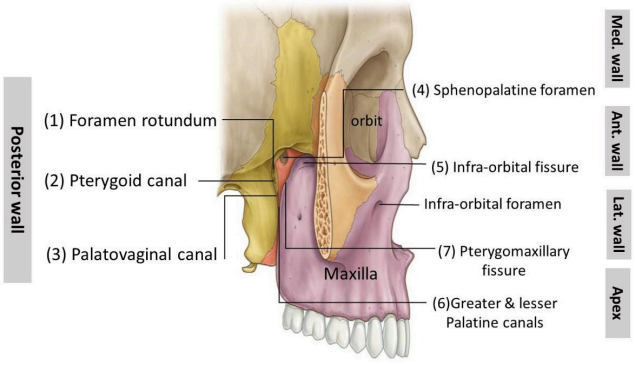
Internal foramina of the skull and their contents

Gray's Atlas of Anatomy, 2nd ed., p. 588

05 Pterygopalatine fossa

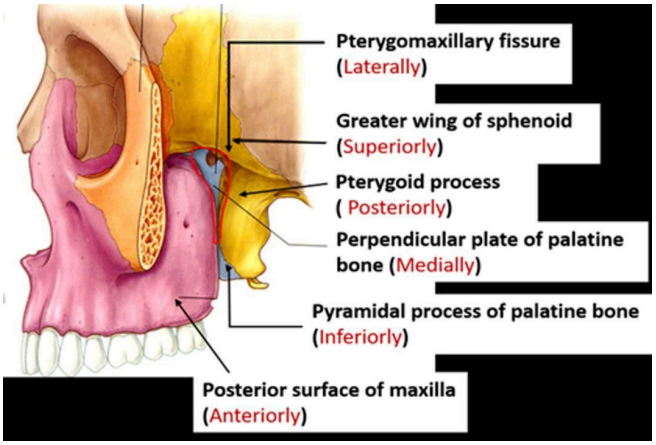
- Small inverted pyramidal space behind the maxilla. **Roof:** body of sphenoid. **Front:** maxilla. **Back:** pterygoid plates. **Medial:** perpendicular plate of the palatine bone.
- **Contents:** third part of the maxillary artery, the maxillary nerve (V2) and the **pterygopalatine (sphenopalatine) ganglion**.
- **Communications:** orbit (inferior orbital fissure), middle cranial fossa (foramen rotundum), foramen lacerum (vidian canal), nasopharynx (palatovaginal canal), nasal cavity (**sphenopalatine foramen**), infratemporal fossa (**pterygomaxillary fissure**), oral cavity (greater and lesser palatine canals).

CLINICAL PEARL Juvenile angiofibroma starts at the sphenopalatine foramen and spreads through these communications. Widening of the pterygopalatine fossa pushes the back wall of the maxilla forward (Holman–Miller sign).



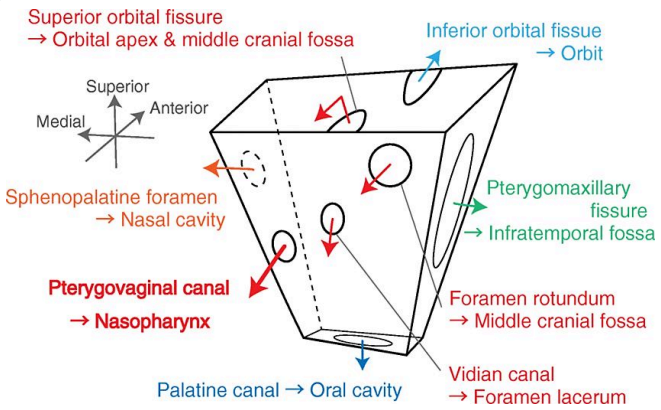
Pterygopalatine fossa and its openings

Dr. Sameh Doss handout, p. 11



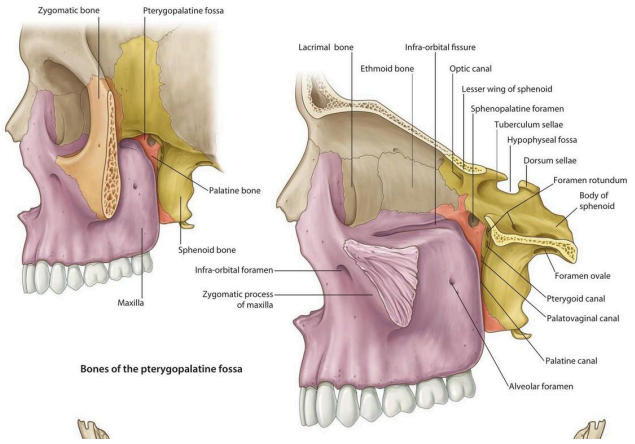
Boundaries of the pterygopalatine fossa

ENT Steps Anatomy, p. 160



Communications of the pterygopalatine fossa

ENT Steps Anatomy, p. 161



Bones of the pterygopalatine fossa

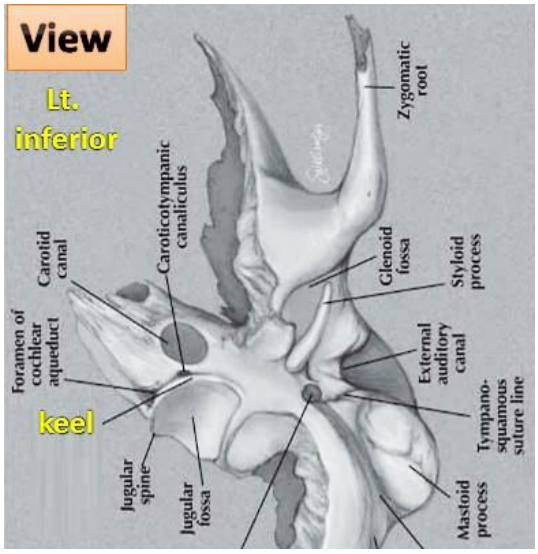
Gray's Atlas of Anatomy, 2nd ed., p. 534

06 Radiological anatomy: On imaging

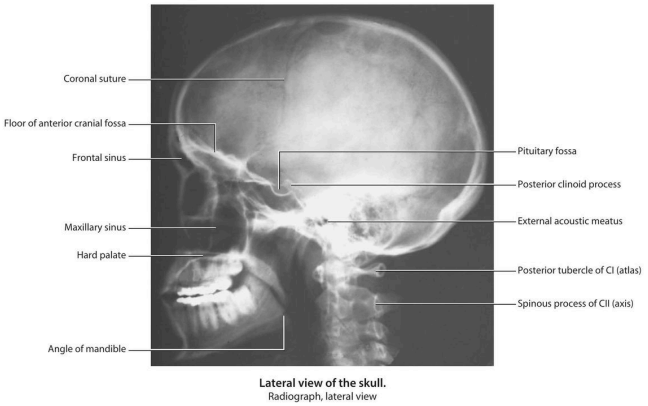
- **HRCT temporal bone** (axial and coronal, about 0.6 mm slices, bone window) is the standard for bony anatomy; **MRI with gadolinium** (T1, T2/CISS) shows the IAC, cochlear fluid, nerves and soft tissue.
- On axial CT, find in order from top: superior canal, then the IAC with the vestibule and lateral canal, then the cochlea and the tympanic facial nerve, then the jugular bulb and carotid canal.

- **Look for:** tegmen thickness and dehiscence, sigmoid sinus position, high or dehiscent jugular bulb, carotid canal dehiscence, superior canal dehiscence (Pöschl view).

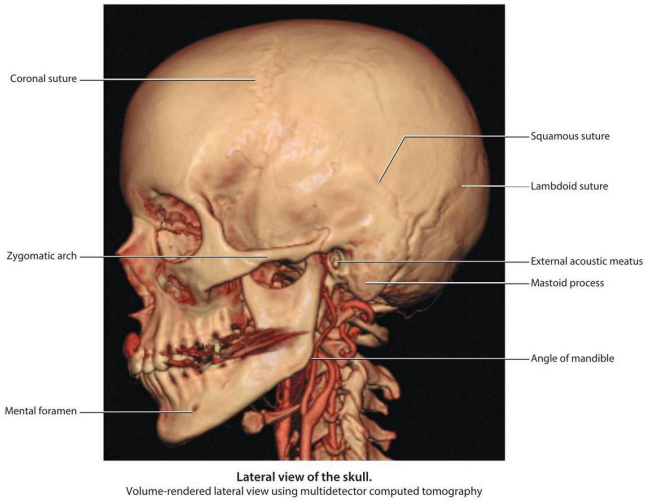
> **CLINICAL PEARL** A high-riding jugular bulb above the round window level can be injured during cochlear implantation or myringotomy.



Inferior surface in detail with the keel between carotid and jugular
ENT Steps Anatomy, p. 24



Lateral skull radiograph: pituitary fossa, sinuses, external acoustic meatus
Gray's Atlas of Anatomy, 2nd ed., p. 481



Lateral view of the skull: volume-rendered CT
Gray's Atlas of Anatomy, 2nd ed., p. 481

07 Surgical anatomy: In theatre

- Skull base approaches are planned around the **ICA, jugular bulb, facial nerve, labyrinth and dura** seen on CT.
- The **arcuate eminence** marks the superior canal only roughly; in the middle fossa approach the greater petrosal nerve and the IAC are safer landmarks.
- **Infratemporal fossa approach (Fisch A)** permanently reroutes the facial nerve anteriorly to reach the jugular foramen.

Tables

Skull base foramina and their contents

Foramen	Bone / fossa	Contents
Cribriform plate	Ethmoid · anterior fossa	Olfactory nerves (I), anterior ethmoidal nerve and vessels
Optic canal	Lesser wing of sphenoid	Optic nerve (II), ophthalmic artery
Superior orbital fissure	Between greater and lesser wings	III, IV, VI, V1 (lacrimal, frontal, nasociliary), superior ophthalmic vein
Foramen rotundum	Greater wing · middle fossa	Maxillary nerve (V2)
Foramen ovale	Greater wing · middle fossa	V3, accessory meningeal artery, lesser petrosal nerve, emissary vein (MALE)
Foramen spinosum	Greater wing · middle fossa	Middle meningeal artery and vein, meningeal branch of V3
Foramen lacerum	Between sphenoid, petrous and occipital	Cartilage; ICA crosses its upper part; greater + deep petrosal form the vidian nerve
Internal acoustic meatus	Petrous · posterior fossa	VII, nervus intermedius, VIII, labyrinthine artery and vein
Jugular foramen	Petrous + occipital · posterior fossa	IX, X, XI, IJV (sigmoid sinus), inferior petrosal sinus
Hypoglossal canal	Occipital	XII, meningeal branch of ascending pharyngeal artery
Stylomastoid foramen	Temporal	Facial nerve (VII), stylomastoid artery
Petrotympanic fissure	Temporal	Chorda tympani, anterior tympanic artery
Foramen magnum	Occipital	Medulla, vertebral arteries, spinal roots of XI

Jugular foramen syndromes: Vernet IX–XI, Collet–Sicard IX–XII, Villaret IX–XII + sympathetic chain.

Pharyngeal arch derivatives

Arch	Nerve	Skeletal	Muscles
1st	V3	Malleus (head), incus (body), Meckel's cartilage, sphenomandibular ligament	Mastication, tensor tympani, tensor veli palatini, mylohyoid, anterior digastric
2nd	VII	Stapes (superstructure), styloid, stylohyoid ligament, lesser horn and upper hyoid body	Facial expression, stapedius, stylohyoid, posterior digastric
3rd	IX	Greater horn and lower hyoid body	Stylopharyngeus
4th	X (superior laryngeal)	Thyroid cartilage	Cricothyroid, pharyngeal constrictors, levator veli palatini
6th	X (recurrent laryngeal)	Cricoid, arytenoid, corniculate	Intrinsic laryngeal muscles (except cricothyroid)

Pouches: 1st → middle ear and ET; 2nd → tonsillar fossa; 3rd → inferior parathyroid and thymus; 4th → superior parathyroid and C cells.

Cranial fossae relations

Fossa	Floor formed by	ENT relations
Anterior	Orbital plates of frontal, cribriform plate, lesser wings	Frontal sinus, ethmoid roof, olfactory bulbs

Fossa	Floor formed by	ENT relations
Middle	Greater wings of sphenoid, squamous and petrous temporal	Tegmen, sphenoid sinus, cavernous sinus, foramina rotundum, ovale, spinosum
Posterior	Occipital bone, posterior petrous surface	IAM, jugular foramen, sigmoid sinus, cerebellum (mastoid relation)

M Mnemonics

Foramen ovale contents

MALE

Mandibular nerve, Accessory meningeal artery, Lesser petrosal nerve, Emissary vein

Superior orbital fissure

Live Frankly To See No Insult At all

Lacrimal, Frontal, Trochlear, Superior division of III, Nasociliary, Inferior division of III, Abducens (plus the superior ophthalmic vein)

Q Rapid recall

1. **Name the five parts of the temporal bone.** Squamous, petrous, mastoid, tympanic and styloid process.
2. **What is the keel of the temporal bone and what passes through it?** The caroticojugular crest between the carotid canal and jugular fossa; the tympanic canaliculus in it carries Jacobson's nerve (IX).
3. **Contents of the foramen ovale?** MALE: Mandibular nerve, Accessory meningeal artery, Lesser petrosal nerve, Emissary vein.
4. **Contents of the jugular foramen?** IX, X, XI, the IJV (from the sigmoid sinus) and the inferior petrosal sinus.
5. **Triad of Gradenigo's syndrome?** Ear discharge, retro-orbital pain (V) and lateral rectus palsy (VI) from petrous apicitis.
6. **Which cranial nerves are involved in Vernet, Collet–Sicard and Villaret syndromes?** Vernet IX–XI; Collet–Sicard IX–XII; Villaret IX–XII plus cervical sympathetic (Horner's).
7. **Three muscles and two ligaments attached to the styloid process?** Stylohyoid, styloglossus, stylopharyngeus; stylohyoid and stylomandibular ligaments.
8. **Through which fissure does the chorda tympani leave the skull?** Petrotympanic (Glaserian) fissure.
9. **What lies medially on the anterior surface of the petrous bone?** The arcuate eminence, raised by the superior semicircular canal.
10. **Where is the trigeminal impression?** At the apex of the anterior petrous surface, for the trigeminal ganglion in Meckel's cave.
11. **Which petrous surface carries the internal acoustic meatus?** The posterior surface (front of the posterior cranial fossa).
12. **What is the subarcuate fossa?** Depression superolateral to the IAM, under the arcuate eminence, carrying the subarcuate artery.
13. **Where does the cochlear aqueduct open externally?** At the upper limit of the jugular foramen on the inferior petrous surface.
14. **Which part of the temporal bone forms most of the bony EAC?** The tympanic part (the squamous part forms its roof).
15. **Two sutures in the bony EAC?** Tympanosquamous in front, tympanomastoid behind.
16. **Where does the posterior belly of digastric arise?** Digastric groove (mastoid notch) on the medial side of the mastoid.
17. **Contents of the internal acoustic meatus?** Facial nerve, nervus intermedius, vestibulocochlear nerve, labyrinthine artery and vein.
18. **Boundaries of the pterygopalatine fossa?** Roof: sphenoid body. Front: maxilla. Back: pterygoid plates. Medial: perpendicular plate of palatine.

19. **What passes through the foramen lacerum in life?** Only small structures pass through its cartilage (meningeal branch of ascending pharyngeal artery, emissary veins); the ICA crosses its upper part and the vidian nerve forms there.
20. **What separates the middle from the posterior cranial fossa?** The petrous ridge of the temporal bone (with the tentorium attached).

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.